



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE (Advanced)

PRE - SCORE

08-10-2023

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TEAS]

ANSWER KEY

PAPER-1 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	A	C	A	C		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A,B,C	A,B	A,B,D	A,C	B,C,D	A,B,D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	30.00	48.00	20.00	10.00	1.25	1.25
SECTION-II (ii)	Q.	7	8	9			
	A.	14	80	7			

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4		
	A.	D	A	D	D		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A,B	B,C	A,B,D	C,D	A,B	A,B,C
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	100.00	0.25	3.00	1.88	2.00	5.00
SECTION-II (ii)	Q.	7	8	9			
	A.	120	32	2			

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	C	B	A	C		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	B,C	A,D	A,B,D	B,C,D	A,B,C	B,C
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	4.00	5.00	3.00	3.00	7.00	3.00
SECTION-II (ii)	Q.	7	8	9			
	A.	8	3	5			

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A)

Along x-axis $\vec{v}_i = 2\vec{v}_m - \vec{v}_0 = 2\hat{i}$ in y-z plane $\vec{v}_i = \vec{v}_0 = -3\hat{j} + 4\hat{k}$

2. Ans (C)

$$1.8 = \frac{f}{f+10}$$

$$1.8f + 18 = f$$

$$18 = -0.8f$$

$$f = -22.5\text{cm}$$

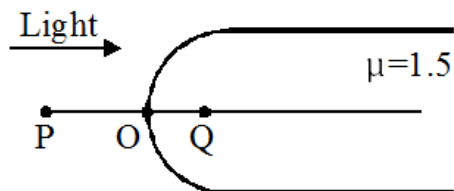
In second case , $u = -50$ $\Rightarrow$ object is beyond C. $\Rightarrow$ image is inverted and diminished.

3. Ans (A)

$$v = OQ, u = OP = -2OQ$$

Equation of refraction at curved surface

with $OP = 2OQ$



$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R} \Rightarrow \frac{1.5}{OQ} - \frac{1}{-2OQ} = \frac{1.5 - 1}{R}$$

$$\Rightarrow \frac{2}{OQ} = \frac{1}{2R} \Rightarrow OQ = 4R \quad \text{therefore } OP$$

$$= 2OQ = 2 \times 4R = 8R$$

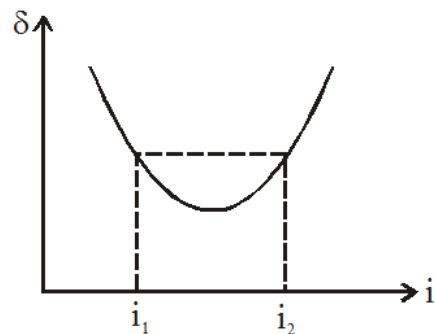
4. Ans (C)

$$\frac{\mu_2}{f_2} - \frac{\mu_1}{\infty} = \frac{\mu_2 - \mu}{-20} + \frac{\mu - \mu_1}{10}$$

PART-1 : PHYSICS

SECTION-I (ii)

5. Ans (A,B,C)



For two different values of incidence we have a single amount of deviation.

(other than minimum deviation).

If we take i_1 as incidence then i_2 will act as emergence and vice-versa.

6. Ans (A,B)

Case : I

$$\frac{h}{T} = v_s = \frac{dH}{dt}$$

$$NS = 2H \left(1 - \frac{1}{n} \right)$$

$$v_I = \frac{2dH}{dt} \left(1 - \frac{1}{n} \right) = \frac{2h}{T} \left(1 - \frac{1}{n} \right)$$

Case : 2

$$NS = (H - h_I) \left(1 - \frac{1}{2H} \right)$$

$$V_I = \frac{dH}{dt} \left(1 - \frac{1}{n} \right)$$

$$= \frac{h}{T} \left(1 - \frac{1}{n} \right)$$

7. Ans (A,B,D)

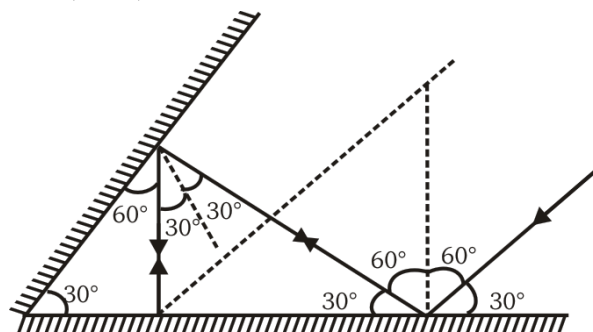
For angle of incidence of 60° light emerges $\Rightarrow$ all rays having incident angle $> 60^\circ$ will emerge.

For angle of incidence of 50° , light suffers TIR

$\Rightarrow$ all rays having angle of incidence $< 50^\circ$ will suffer TIR.

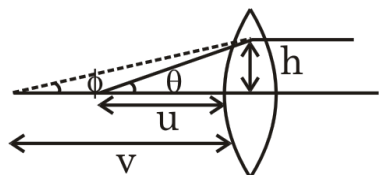
For rays between 50° to 60° as angle of incidence, nothing can be commented.

8. Ans (A,C)

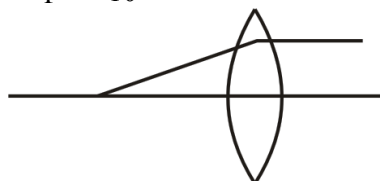


9. Ans (B,C,D)

$$d_{\min} = 4f = 40 \text{ cm.}$$

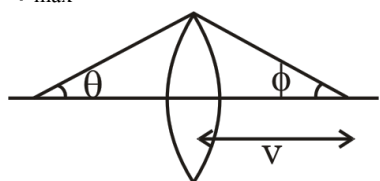


$$\begin{aligned} \delta &= \theta - \phi \\ &= \frac{h}{-u} + \frac{h}{v} \\ &= \frac{h}{f} = \frac{0.5}{10} \end{aligned}$$



$$\tan \phi_{\max} = \frac{2h}{v} = \infty$$

$$\phi_{\max} = \pi$$



$$\phi = \frac{h}{v}$$

$$\phi_{\max} \text{ when } v \text{ is min}$$

$$\Rightarrow v = f$$

$$\phi_{\max} = \frac{0.5}{10}$$

$$2\phi = 0.1 \text{ rad.}$$

10. Ans (A,B,D)

$$f_1 = \frac{a}{2} \rightarrow \text{erect}$$

$$\frac{1}{f_2} = -2(1.5 - 1) \left(\frac{1}{a} + \frac{1}{a} \right) \pm \frac{2}{a}$$

$$= -\frac{4}{a} \Rightarrow f_2 = -\frac{a}{4} \rightarrow \text{inverted}$$

$$\text{For real image } |u| > |2f| = \frac{a}{2}$$

PART-1 : PHYSICS

SECTION-II (i)

1. Ans (30.00)

$$m = -\frac{f}{x_0} = \frac{+20}{+10} = 2$$

$$m' = -2 = \frac{20}{x} \Rightarrow -10 = x$$

$$\Rightarrow \text{distance} = 30 \text{ cm}$$

2. Ans (48.00)

$$v_1 = -m^2 v_0 = -(2)^2 v_0 = -4 \times 12 \hat{i} = -48 \hat{i}$$

3. Ans (20.00)

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u} = \frac{1}{30} + \frac{1}{60}$$

$$\Rightarrow \frac{1}{f} = \frac{1}{20} \Rightarrow f = 20 \text{ cm}$$

Focal length of convex mirror

$$f = \frac{1}{2} (30 - 10) = 10 \text{ cm}$$

4. Ans (10.00)

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u} = \frac{1}{30} + \frac{1}{60}$$

$$\Rightarrow \frac{1}{f} = \frac{1}{20} \Rightarrow f = 20 \text{ cm}$$

Focal length of convex mirror

$$f = \frac{1}{2} (30 - 10) = 10 \text{ cm}$$

5. Ans (1.25)

$$\frac{1}{f} = (\mu_1 - 1) \left(\frac{1}{R_1} + \frac{1}{R_2} \right) + (\mu_2 - 1) \left(\frac{1}{-R_2} + \frac{1}{R_0} \right)$$

Achromatic $\Rightarrow$

$$\beta_1 \left(\frac{1}{R_1} + \frac{1}{R_2} \right) + \beta_1 \left(-\frac{1}{R_2} + \frac{1}{R_0} \right) = 0$$

$$0.05 \left(\frac{1}{10} + \frac{1}{R_2} \right) + 0.06 \left(-\frac{1}{R_2} + \frac{1}{20} \right) = 0$$

$$\frac{0.16}{20} = \frac{0.01}{R_2}$$

$$R_2 = \frac{20}{16} = \frac{5}{4} = 1.25 \text{ cm}$$

6. Ans (1.25)

$$\frac{1}{f_1} + \frac{1}{f_2} = 0$$

$$(n_{01} + \beta_1 (\lambda_1 - \lambda_0) - 1) \left(\frac{1}{R_1} + \frac{1}{R_2} \right) +$$

$$(n_{02} + \beta_2 (\lambda_1 - \lambda_0) - 1) \left(\frac{1}{-R_2} + \frac{1}{R_0} \right) = 0$$

$$\Rightarrow (1.5 + 0.05 \times 20 - 1) \left[\frac{1}{10} + \frac{1}{R_2} \right] + \{1.6 +$$

$$0.06 \times 20 - 1 \left[-\frac{1}{R_2} + \frac{1}{20} \right] = 0$$

$$\frac{1.5}{10} + \frac{1.5}{R_2} - \frac{1.8}{R_2} + \frac{1.8}{20} = 0$$

$$\Rightarrow \frac{4.8}{20} = \frac{0.3}{R_2} \Rightarrow R_2 = 1.25 \text{ cm}$$

PART-1 : PHYSICS

SECTION-II (ii)

7. Ans (14)

$$u = - \left(4 + \frac{4}{2} \right) \times 2 = -12 \text{ cm}$$

$$\frac{1}{v} + \frac{1}{-12} = \frac{1}{-8}$$

$$\Rightarrow v = -24 \text{ cm.}$$

for final image (refraction through plane surface) we can write.

$$24 = 2 \left(\frac{x}{1} + \frac{4}{2} \right)$$

$$\Rightarrow 24 = 2x + 4$$

$$x = 10 \text{ cm}$$

so distance from P is 14 cm.

8. Ans (80)

+ 80 cm, convergent achromatic lens

For achromatic combination of lens,

$$\frac{\omega_1}{f_1} + \frac{\omega_2}{f_2} = 0$$

$$\omega = \frac{\mu_v - \mu_r}{\left(\frac{\mu_v + \mu_r}{2} \right) - 1}$$

$$\text{Crown glass } \omega_1 = \frac{1.525 - 1.515}{\left(\frac{1.525 + 1.515}{2} \right) - 1} = \frac{0.010}{0.520} = \frac{1}{52}$$

$$\text{Flint glass } \omega_2 = \frac{1.655 - 1.645}{\left(\frac{1.655 + 1.645}{2} \right) - 1} = \frac{0.010}{0.650} = \frac{1}{65}$$

$$f = -20 \text{ cm}$$

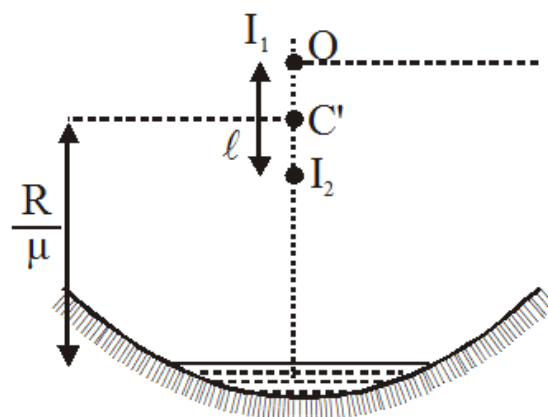
Focal length of combination.

$$\frac{1}{F_{eq}} = \frac{1}{f_1} + \frac{1}{f_2}$$

$$= \frac{1}{16} - \frac{1}{20}$$

$$F_{eq} = +80 \text{ cm converging.}$$

9. Ans (7)



On pouring water the new center of curvature is

at R/μ .

$$\frac{1}{v} + \frac{1}{-R} = \frac{2\mu}{-R}$$

$$\therefore v = \frac{R}{1 - 2\mu} \text{ for } I_2; v = -5$$

$$\mu = 1.4$$

PART-2 : CHEMISTRY

SECTION-II (ii)

7. Ans (120)

$$P = 6$$

$$Q = 4$$

$$R = 5$$

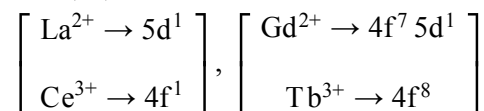
$$P \times Q \times R = 6 \times 4 \times 5 = 120$$

8. Ans (32)

$$X = 8$$

$$Y = 4$$

9. Ans (2)



An important take home lesson ! Except the

above two pairs, all other pairs have same

outermost E.C.

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (C)**

$$t = r + b \quad t \leq 2024$$

$$\frac{{}^r C_2}{{}^t C_2} + \frac{{}^b C_2}{{}^t C_2} = \frac{1}{2}$$

$$\frac{r(r-1)}{t(t-1)} + \frac{b(b-1)}{t(t-1)} = \frac{1}{2}$$

$$2(r(r-1) + (t-r)(t-r-1)) = t(t-1)$$

$$4r^2 - 4rt + t(t-1) = 0$$

$$r = \frac{4t \pm \sqrt{16t^2 - 16(t^2 - t)}}{2(4)}$$

$$r = \frac{t \pm \sqrt{t}}{2}$$

$$\text{Now } t \leq 2024 \quad r \in \mathbb{N}.$$

$$\Rightarrow r \max_{t=1936} = 990$$

2. **Ans (B)**

Let $(1 + 2p, 1 + 3p, -3 + 2p)$ be a point on the first line and $(k + q, -1 + 5q, -q)$ be a point on the second line. If they are co-ordinates of point of intersection then

$$-3 + 2p = -q \text{ and } 1 + 3p = -1 + 5q$$

$$1 + 2p = k + q$$

$$\Rightarrow p = q = 1 \text{ and } k = 2$$

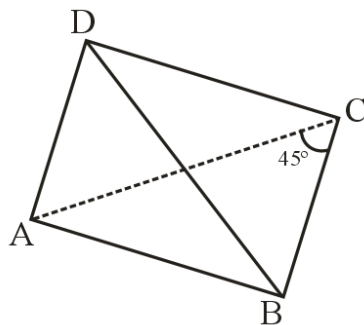
Point of intersection will be $(3, 4, -1)$

3. **Ans (A)**

$$\frac{1}{3} AD \left(\frac{1}{2} BC \cdot AC \cdot \sin 45^\circ \right) \geq V_{ABCD} = \frac{1}{6}$$

$$\text{So, } AD \cdot BC \cdot$$

$$\frac{AC}{\sqrt{2}} \geq 1$$



$$3 = AD + BC + \frac{AC}{\sqrt{2}} \geq 3 \sqrt{AD \cdot BC \cdot \frac{AC}{\sqrt{2}}} \geq 3$$

$$\text{Equality holds if and only if } AD = BC = \frac{AC}{\sqrt{2}} = 1$$

So AD is perpendicular to the face ABC.

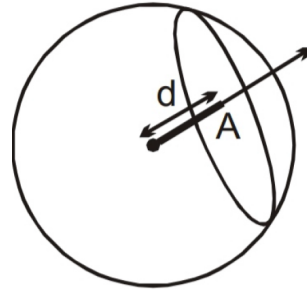
$$DC = \sqrt{AD^2 + AC^2} = \sqrt{3}$$

4. **Ans (C)**

d = distance of $(0,0,0)$ from $x + y + z$

$$= 3\sqrt{3} = \left| \frac{3\sqrt{3}}{\sqrt{3}} \right| = 3$$

$$A(\vec{a}) = 3 \left(\frac{\hat{i} + \hat{j} + \hat{k}}{\sqrt{3}} \right)$$



PART-3 : MATHEMATICS

SECTION-I (ii)

5. **Ans (B,C)**

$$\vec{r} = x(\vec{b} \times \vec{c}) + y(\vec{c} \times \vec{a}) + z(\vec{a} \times \vec{b})$$

$$\vec{r} \cdot \vec{a} = p_1 = x[\vec{a} \vec{b} \vec{c}]$$

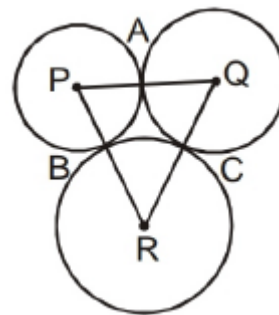
6. **Ans (A,D)**

$$PA = PB = 1$$

$$QA = QC = 2$$

$$RB = RC = 3$$

$$\Rightarrow PQ = 3; \quad QR = 5; \quad PR = 4$$



$\Rightarrow \Delta PQR$ will be a right angled triangle with $\angle P = 90^\circ$;

$$\angle Q = \sin^{-1} \frac{4}{5} \text{ and } \angle R = \sin^{-1} \frac{3}{5}$$

Area of ΔABC

$$= A(\Delta PQR) - A(\Delta PAB) - A(\Delta QAC) - A(\Delta RBC)$$

$$= 6 - \left(\frac{1}{2} \right) - \frac{8}{5} - \frac{27}{10} = \frac{60 - 5 - 16 - 27}{10} = \frac{6}{5}$$

Circum-radius of $\Delta ABC = 1$

7. Ans (A,B,D)

Let $P(A) = a, P(B) = b, P(C) = c$

$$\Rightarrow P(A \cup B \cup C) = P(A) + P(B) + P(C) -$$

$$P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$$

$$\Rightarrow 1 = a + b + c - ab - bc - ca + abc$$

$$\Rightarrow 1 - (a + b + c) + (ab + bc + ca) - abc = 0$$

$$\Rightarrow (1 - a)(1 - b)(1 - c) = 0$$

$\Rightarrow$ at least one among a, b, c is 1.

8. Ans (B,C,D)

$$\begin{bmatrix} \vec{a} & \vec{b} & \vec{c} \end{bmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 3 & 1 \end{vmatrix} = -4 + 1 + 5 = 2$$

$$\vec{c} \times \vec{b} = \vec{d} \times \vec{b} \Rightarrow (\vec{c} - \vec{d}) \times \vec{b} = 0 \Rightarrow \vec{b}, \vec{c} \text{ and } \vec{d}$$

are coplanar.

$$\Rightarrow \vec{c} = \vec{d} + \lambda \vec{b} \Rightarrow \vec{c} \cdot \vec{c} = \vec{d} \cdot \vec{c} + 0 = 14$$

9. Ans (A,B,C)

$$P(A) \times P(B) = \frac{1}{8} \text{ and } (1 - P(A)) \times (1 - P(B)) = \frac{3}{8}$$

Treat these equations as $xy = \frac{1}{8}$ and $(1 - x)(1 - y) = \frac{3}{8}$

$$= \frac{3}{8} \text{ and solve}$$

$$P(A) = \frac{1}{2} \text{ or } \left(\frac{1}{4}\right) \text{ and } P(B) = \frac{1}{4} \text{ or } \left(\frac{1}{2}\right)$$

10. Ans (B,C)

Fix n . Let $Q(n) = P(m/n) \therefore Q(0) = 0, Q(m) = 1$

$$Q(n) = \frac{Q(n-1)}{2} + \frac{Q(n+1)}{2}$$

$$\Rightarrow Q(n) - Q(n-1) = Q(n+1) - Q(n) \Rightarrow A.P.$$

$$Q(n) - Q(n-1) = Q(n-1) - Q(n-2) = \dots =$$

$$Q(1) - Q(0)$$

$$\therefore Q(n) = nQ(1). \text{ Also } Q(m) = mQ(1) \therefore Q(1)$$

$$= \frac{1}{m}$$

$$\therefore Q(n) = \frac{n}{m}$$

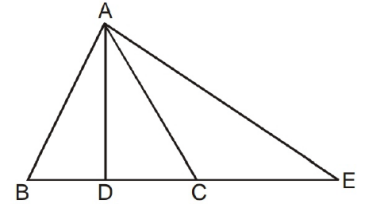
PART-3 : MATHEMATICS

SECTION-II (i)

1. Ans (4.00)

$\angle EAD = \alpha$ and $\angle CAD = \beta$

$\tan(\alpha - \beta), \tan \alpha$ and $\tan(\alpha + \beta)$ are in G.P.



$$\Rightarrow \tan^2 \alpha = \frac{\sin(\alpha - \beta) \sin(\alpha + \beta)}{\cos(\alpha - \beta) \cos(\alpha + \beta)} = \frac{\sin^2 \alpha - \sin^2 \beta}{\cos^2 \alpha - \sin^2 \beta}$$

$$\Rightarrow \sin^2 \alpha \cos^2 \alpha - \sin^2 \alpha \sin^2 \beta = \cos^2 \alpha \sin^2 \alpha - \cos^2 \alpha \sin^2 \beta$$

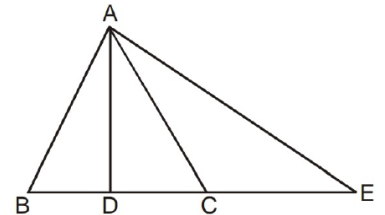
$$\Rightarrow \sin^2 \beta = 0 \text{ or } \tan^2 \alpha = 1 \Rightarrow \alpha = \frac{\pi}{4} (\beta \neq 0)$$

$$\Rightarrow AD = AE \cos \frac{\pi}{4} = 5\sqrt{2}$$

2. Ans (5.00)

$\angle EAD = \alpha$ and $\angle CAD = \beta$

$\tan(\alpha - \beta), \tan \alpha$ and $\tan(\alpha + \beta)$ are in G.P.



$$\Rightarrow \tan^2 \alpha = \frac{\sin(\alpha - \beta) \sin(\alpha + \beta)}{\cos(\alpha - \beta) \cos(\alpha + \beta)} = \frac{\sin^2 \alpha - \sin^2 \beta}{\cos^2 \alpha - \sin^2 \beta}$$

$\Rightarrow$

$$\sin^2 \alpha \cos^2 \alpha - \sin^2 \alpha \sin^2 \beta = \cos^2 \alpha \sin^2 \alpha - \cos^2 \alpha \sin^2 \beta$$

$$\Rightarrow \sin^2 \beta = 0 \text{ or } \tan^2 \alpha = 1 \Rightarrow \alpha = \frac{\pi}{4} (\beta \neq 0)$$

$$\Rightarrow AD = AE \cos \frac{\pi}{4} = 5\sqrt{2}$$

3. Ans (3.00)

$$A(3a + 7, 2a + 5, a + 3)$$

$$B(2b + 1, 4b - 1, 3b - 1)$$

$$\frac{3a - 2b + 6}{2} = \frac{2a - 4b + 6}{2} = a - 3b + 4$$

$$\Rightarrow a + 2b = 0 \text{ and } a - 2b + 3 = a - 3b + 4$$

$$a = -2 \quad b = 1$$

$$\Rightarrow A(1, 1, 1)$$

$$B(3, 3, 2)$$

$$AB = \sqrt{2^2 + 2^2 + 1^2} = 3$$

4. **Ans (3.00)**

$$A(3a + 7, 2a + 5, a + 3)$$

$$B(2b + 1, 4b - 1, 3b - 1)$$

$$\frac{3a - 2b + 6}{2} = \frac{2a - 4b + 6}{2} = a - 3b + 4$$

$$\Rightarrow a + 2b = 0 \text{ and } a - 2b + 3 = a - 3b + 4$$

$$a = -2 \quad b = 1$$

$$\Rightarrow A(1, 1, 1)$$

$$B(3, 3, 2)$$

$$AB = \sqrt{2^2 + 2^2 + 1^2} = 3$$

5. **Ans (7.00)**

$$P_{aA} = \frac{1}{3} + \frac{1}{3}P_{bA} + \frac{1}{3}P_{dA}$$

$$P_{bA} = P_{dA} \text{ and } P_{cA} = \frac{2}{3}P_{bA}$$

$$P_{bA} = \frac{1}{3}P_{aA} + \frac{1}{3}P_{cA} \Rightarrow P_{bA} = \frac{3}{7}P_{aA}$$

$$\Rightarrow P_{aA} = \frac{7}{15} \quad P_{bA} = \frac{1}{5}$$

6. **Ans (3.00)**

$$P_{aA} = \frac{1}{3} + \frac{1}{3}P_{bA} + \frac{1}{3}P_{dA}$$

$$P_{bA} = P_{dA} \text{ and } P_{cA} = \frac{2}{3}P_{bA}$$

$$P_{bA} = \frac{1}{3}P_{aA} + \frac{1}{3}P_{cA} \Rightarrow P_{bA} = \frac{3}{7}P_{aA}$$

$$\Rightarrow P_{aA} = \frac{7}{5} \quad P_{bA} = \frac{1}{5}$$

PART-3 : MATHEMATICS

SECTION-II (ii)

7. **Ans (8)**

$$\text{Volume of DFAB} = \frac{1}{3} \times \text{Height} \times \text{Base Area}$$

$$= \frac{1}{2}(\text{DE})(\text{area } \triangle OAB) = \frac{1}{3}(6)(4) = 8$$

(Given line is perpendicular to AB and passes through E.)

8. **Ans (3)**

$$|\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}|^2$$

$$= |\vec{a} \times \vec{b}|^2 + |\vec{b} \times \vec{c}|^2 + |\vec{c} \times \vec{a}|^2 + 2(\vec{a} \times \vec{b}) \cdot (\vec{b} \times \vec{c}) + 2(\vec{b} \times \vec{c}) \cdot (\vec{c} \times \vec{a}) + 2(\vec{c} \times \vec{a}) \cdot (\vec{a} \times \vec{b})$$

$$= \frac{3}{4} \times 3 + 2\left(\frac{-1}{4}\right) \times 3 = \frac{3}{4}$$

using

$$|\vec{a} \times \vec{b}| = |\vec{b} \times \vec{c}| = |\vec{c} \times \vec{a}| = \frac{\sqrt{3}}{2}$$

$$(\vec{a} \times \vec{b}) \cdot (\vec{b} \times \vec{c}) = \begin{vmatrix} \vec{a} \cdot \vec{b} & \vec{b} \cdot \vec{b} \\ \vec{a} \cdot \vec{c} & \vec{b} \cdot \vec{c} \end{vmatrix} = \begin{vmatrix} \frac{1}{2} & 1 \\ \frac{1}{2} & \frac{1}{2} \end{vmatrix} = -\frac{1}{4}$$

9. **Ans (5)**

$$\text{Let } a = \frac{{}^3C_2 \cdot {}^5C_2}{{}^8C_4} \cdot \frac{{}^2C_2}{{}^4C_2}; b = \frac{{}^3C_3 \cdot {}^5C_1}{{}^8C_4} \cdot \frac{{}^3C_2}{{}^4C_2}$$

$$\frac{p}{q} = \frac{a}{a+b} \cdot \frac{2}{4} + \frac{b}{a+b} \cdot \frac{3}{4} = \frac{7}{12}$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE (Advanced)

PRE - SCORE

08-10-2023

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TEAS]

ANSWER KEY

PAPER-2 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,B	B,D	A,B,C	C,D	B,D	B,C,D
SECTION-I (ii)	Q.	7	8	9	10		
	A.	A	C	B	B		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	0.00	124.00	41.00	1.73	0.33	4.00
SECTION-II (ii)	Q.	7	8	9			
	A.	5	2	15			

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,C	A,B,D	B,C	A,B,C,D	A,C	A,B,D
SECTION-I (ii)	Q.	7	8	9	10		
	A.	B	A	D	A		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	62.00	9.00	0.50	13.60	8.00	3.00
SECTION-II (ii)	Q.	7	8	9			
	A.	28	19	7			

PART-3 : MATHEMATICS

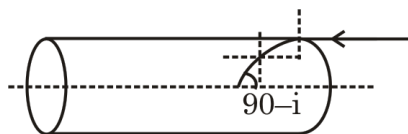
SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,B,C,D	A,B,D	B,C	A,B,D	A,D	A,D
SECTION-I (ii)	Q.	7	8	9	10		
	A.	A	B	C	B		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	31.00	320.00	49.00	330.00	0.25	9.50
SECTION-II (ii)	Q.	7	8	9			
	A.	7	135	2			

HINT – SHEET

PART-1 : PHYSICS

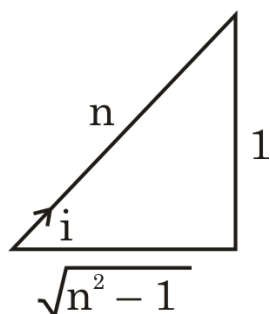
SECTION-I (i)

2. Ans (B,D)



$$n \sin i = 1 \sin 90$$

$$\sin i = \frac{1}{n}$$



$$\sqrt{n^2 - 1} = \cot i = \frac{dr}{dx}$$

$$dx = \frac{dr}{\sqrt{n^2 - 1}} = \frac{dr}{dn} \times \frac{dn}{\sqrt{n^2 - 1}}$$

$$\frac{dn}{dr} = -\frac{(n_0 - 1)}{r_0}$$

$$\int dx = +\frac{r_0}{n_0 - 1} \int \frac{dn}{\sqrt{n^2 - 1}}$$

$$x = +\frac{r_0}{n_0 - 1} \times \ell n \left(n + \sqrt{n^2 - 1} \right) \Big|_1^n$$

$$x = 4r_0 \ell n 2$$

3. Ans (A,B,C)

n for liquid = n for glass / yellow light

but n for liquid < n for glass (red light), so it will deviated towards base, for blue light n (liquid) > n (glass) so it will deviate towards vertex.

5. Ans (B,D)

(i) As the ray incidents normally on face BC, hence it retraces the path.

Thus the deviation of ray = 180° .

$$\begin{aligned} \text{(ii) Required path length} &= 2 \left[\frac{x\sqrt{3}}{2} + (a-x)\frac{\sqrt{3}}{2} \right] \\ &= 2 \left[x\sqrt{3} + \frac{\sqrt{3}}{2}a - \sqrt{3}x \right] = \sqrt{3}a \end{aligned}$$

6. Ans (B,C,D)

If $\mu_2 < \sqrt{3}$; light will show TIR at the first surface i.e. it will return to medium (1) & will not go to medium (3).

If $\mu_2 > \sqrt{3}$, light will show refraction and will enter medium (3) at angle of refraction 90° i.e. deviation = 30° . It will not return to medium (1) again.

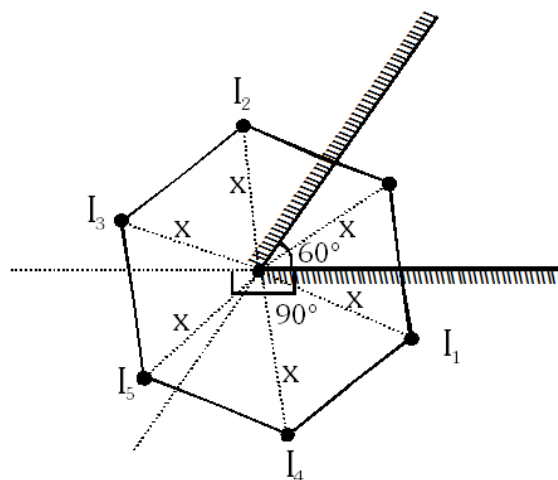
PART-1 : PHYSICS

SECTION-I (ii)

7. Ans (A)

$$\text{Area} \propto x^2$$

8. Ans (C)



$$\text{Total area} = 6 \times \left(\frac{1}{2} x^2 \sin 60^\circ \right) = \frac{3\sqrt{3}x^2}{2}$$

10. **Ans (B)**

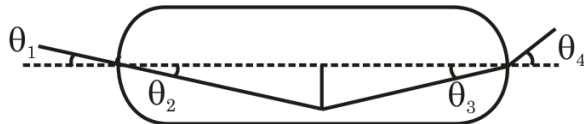
$$\frac{1.5}{v_1} - \frac{1}{\infty} = \frac{1.50 - 1}{10}$$

$$v_1 = 30$$

$$u_2 = -15 \text{ cm}$$

$$\frac{1}{\infty} - \frac{-1.5}{-15} = \frac{1 - 1.5}{R}$$

$$R = -5$$



$$1\theta_1 = 1.5\theta_2 = 1.5 \frac{h}{30}$$

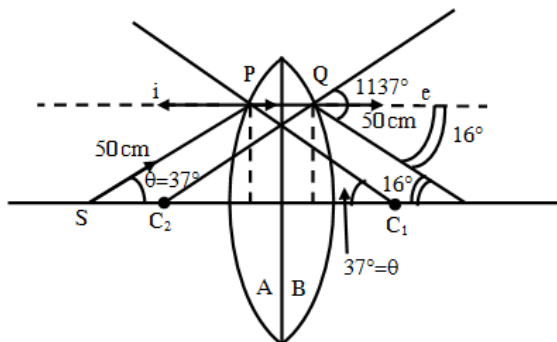
$$1\theta_4 = 1.5\theta_3 = \frac{1.5h}{15}$$

$$\frac{\theta_4}{\theta_1} = \frac{h/10}{h/20} = 2$$

PART-1 : PHYSICS

SECTION-II (i)

1. **Ans (0.00)**



By figure

$$i = \theta + \theta = 2\theta = 74^\circ$$

by shell's law

$$1 \sin 74 = \frac{8}{5} \sin \gamma$$

$$2 \left(\frac{3}{5} \right) \left(\frac{4}{5} \right) = \frac{8}{5} \sin \gamma$$

$$\gamma = 37^\circ$$

So light becomes parallel to principle axis

hence

$$D = \infty$$

$$\Rightarrow \frac{1}{D} = 0$$

2. **Ans (124.00)**

For next surface let angle of emergence is 'e' and

by symmetry angle of incidence = 37°

$$\text{Total deviation} = 53^\circ = (i - r_1) + (e - r_2)$$

$$r_1 = r_2 = 37^\circ \quad i = 74^\circ$$

$$53^\circ = 74^\circ - 37^\circ + e - 37^\circ$$

$$e = 53^\circ$$

Total length - SP + PQ + QI

$$SP = 50 \text{ cm}$$

$$PQ = 20 \text{ cm}$$

$$QI = \frac{30}{\sin 16^\circ}$$

$$\sin 16 = \sin(53 - 37)$$

$$= \frac{4}{5} \cdot \frac{4}{5} - \frac{3}{5} \cdot \frac{3}{5} = \frac{7}{25}$$

$$\text{Total length} = 50 + 20 + \frac{30}{7} \times 25$$

$$= 70 + \frac{750}{7} = \frac{490 + 750}{7}$$

$$\frac{10}{7} Q = \frac{1240}{7} \text{ cm}$$

$$Q = 124$$

3. **Ans (41.00)**

$$\delta = i + e - A$$

$$\delta_{\min} = 2i - A$$

$$60^\circ = 120^\circ - A \Rightarrow A = 60^\circ$$

$$63 = i_1 + 2i_1 - 60^\circ$$

$$i_1 = \frac{123}{3} = 41$$

$$\mu = \frac{\sin \left(\frac{60^\circ + 60^\circ}{2} \right)}{\sin \frac{60^\circ}{2}} = \sqrt{3}$$

4. **Ans (1.73)**

$$\delta = i + e - A$$

$$\delta_{\min} = 2i - A$$

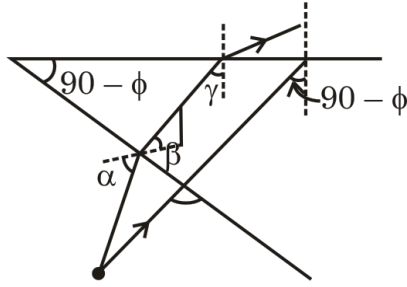
$$60^\circ = 120^\circ - A \Rightarrow A = 60^\circ$$

$$63 = i_1 + 2i_1 - 60^\circ$$

$$i_1 = \frac{123}{3} = 41$$

$$\mu = \frac{\sin \left(\frac{60^\circ + 60^\circ}{2} \right)}{\sin \frac{60^\circ}{2}} = \sqrt{3}$$

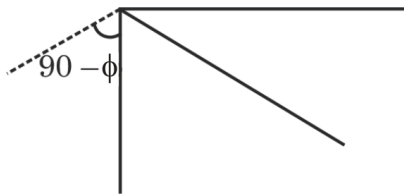
5. Ans (0.33)



As α increases, β increases so γ decreases

$\Rightarrow$ more likely that ray comes out.

γ is min when ray is vertical.



$$i \sin(90 - \phi) = \frac{5}{3} \sin \beta$$

$$\frac{5}{3} \sin(90 - \phi - \beta) = 1$$

$$\cos \phi = \frac{5}{3} \sin \beta$$

$$\frac{5}{3} [\cos \phi \cos \beta - \sin \phi \sin \beta] = 1$$

$$\frac{5}{3} \cos \phi \times \sqrt{1 - \frac{9}{25} \cos^2 \phi} - \sin \phi \cos \phi = 1$$

$$\frac{5}{3} \cos \phi \times \sqrt{1 - \frac{9}{25} \cos^2 \phi} - \sin \phi \cos \phi = 1$$

$$\frac{25}{g} \cos^2 \phi \left(1 - \frac{9}{25} \cos^2 \phi \right) = 1 + 2 \sin \phi \cos \phi$$

$$+ \cos^2 \phi - \cos^4 \phi$$

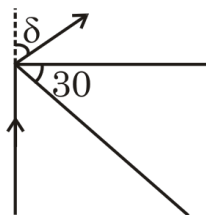
$$\frac{16}{9} \cos^2 \phi = 1 + \sin 2\phi$$

$$\frac{16}{9} = \sec^2 \phi + 2 \tan \phi$$

$$\tan^2 \phi + 2 \tan \phi - \frac{7}{9} = 0$$

$$\tan \phi = \frac{-2 \pm \sqrt{4 + \frac{28}{9}}}{2} = \frac{4}{3} - 1 = \frac{1}{3}$$

6. Ans (4.00)



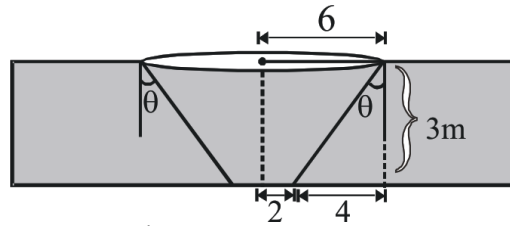
$$\delta = (n - 1)A = \frac{2}{3} \times 3^\circ = 2^\circ$$

$$\text{angle} = 2\delta = 4^\circ$$

PART-1 : PHYSICS

SECTION-II (ii)

8. Ans (2)



$$\theta_C = \sin^{-1} \frac{4}{5}$$

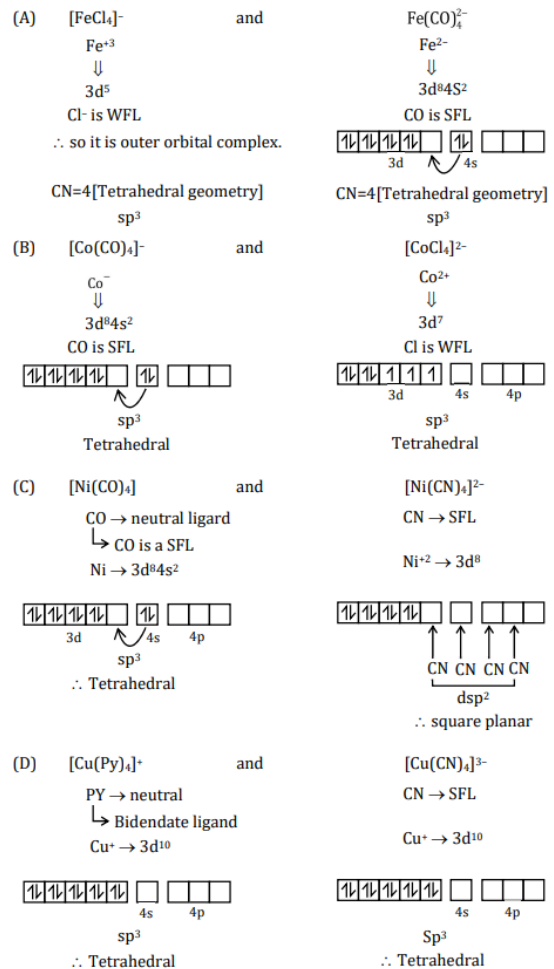
9. Ans (15)

15 cm

PART-2 : CHEMISTRY

SECTION-I (i)

6. Ans (A,B,D)



PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A,B,C,D)

U, N, I², V, E, R, S, T, Y

$$(A) \frac{8! \cdot {}^9C_2 \cdot 1}{\frac{10!}{2!}} = \frac{4}{5}$$

(B) 4 vowels, 6 consonants

$$\frac{6! \cdot {}^7C_4 \cdot \frac{4!}{2!}}{\frac{10!}{2!}} = \frac{1}{6}$$

(C) U, I, I, E, N, V, R, S, T, Y

$$\Rightarrow \frac{7! \cdot \frac{4!}{2!}}{\frac{10!}{2!}} = \frac{1}{30}$$

$$(D) \frac{6}{\downarrow} \quad \frac{5}{\downarrow} = \frac{30 \times \frac{8!}{2!}}{\frac{10!}{2!}} = \frac{1}{3}$$

2. Ans (A,B,D)

$$\vec{p} + \vec{q} + \vec{r} = \vec{0}$$

(A) $\vec{p}, \vec{q}, \vec{r}$ are collinear

$$(B) x = y = z \Rightarrow \frac{x+z}{y} = 2$$

$$(C) (\vec{p} + \vec{q} + \vec{r})^2 = |\vec{p}|^2 + |\vec{q}|^2 + |\vec{r}|^2 + 2\vec{p} \cdot \vec{q} = 0$$

$$(D) (\vec{p} + \vec{r})^2 = (-\vec{q})^2$$

3. Ans (B,C)

$$AM = \frac{\frac{a}{3} + \frac{b}{4} + \frac{c}{3}}{3} = \frac{1}{3} = GM$$

$$\Rightarrow \frac{a}{3} = \frac{b}{4} = \frac{c}{3} = k$$

$$\Rightarrow a = 3k, b = 4k, c = 3k$$

$$abc = \frac{4}{3} \Rightarrow k = \frac{1}{3}$$

$$\therefore a = 1, b = \frac{4}{3}, c = 1$$

4. Ans (A,B,D)

$$\text{On solving } \frac{x}{2} = \frac{y-2}{2} = \frac{z-1}{0} \text{ and } x = 0$$

$$P(0, 2, 1)$$

Image of a point A(2, 4, 1) on the line in $x = 0$ is

$$A'(-2, 4, 1)$$

$$\therefore A'P : \frac{x-0}{2} = \frac{y-2}{-2} = \frac{z-1}{0}$$

On solving A'P and $x + 2y = 0$, get Q(4, -2, 1)

Image of P(0, 2, 1) in plane $x + 2y = 0$ is

$$P' \left(-\frac{8}{5}, -\frac{6}{5}, 1 \right)$$

$$\therefore P'Q : \frac{x-4}{7} = \frac{y+2}{-1} = \frac{z-1}{0}$$

On solving P'Q and incident ray, R(-3, -1, 1)

5. Ans (A,D)

Let E_i = Bag contains i black and $12 - i$ white balls.

$$P(E_i) = \frac{1}{13} \quad (i = 0, 1, 2, \dots, 12)$$

A : All four balls are black

$$P(A) = \sum_{i=0}^{12} P(E_i) \cdot P(A/E_i) = \frac{1}{13} \left(\frac{{}^4C_4 + {}^5C_4 + \dots + {}^{12}C_4}{{}^{12}C_4} \right)$$

$$P(A) = \frac{{}^{13}C_5}{{}^{13}C_4} = \frac{1}{5}$$

Now B : 2 balls are black and 2 are white

$$P(B) = \sum_{i=0}^{12} P(E_i) \cdot P(B/E_i)$$

$$P(B) = \frac{1}{13} \left[\frac{{}^2C_2 \cdot {}^{10}C_2 + {}^3C_2 \cdot {}^9C_2 + {}^4C_2 \cdot {}^8C_2 + \dots + {}^{10}C_2 \cdot {}^2C_2}{{}^{12}C_4} \right]$$

$$P(B) = \frac{1}{5}$$

6. Ans (A,D)

(A) Magnitude and angle same for regular tetrahedron

$$(B) -(\vec{a} \times \vec{b}) \cdot \{\vec{b} + \vec{a} \times \vec{b}\} = -(\vec{a} \times \vec{b})^2 = -\frac{3}{4}$$

$$(D) \vec{b} \times \vec{c} = \vec{a}$$

$$\vec{b} \times (\vec{a} \times \vec{b}) = \vec{a}$$

$$(\vec{b} \cdot \vec{b}) \vec{a} - (\vec{b} \cdot \vec{a}) \vec{b} = \vec{a}$$

$$\Rightarrow \vec{a} \cdot \vec{b} = 0$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. **Ans (A)**

Prob = P(exactly one 4) + P(exactly two 4) + P(exactly three 4)

$$\text{Prob.} = \frac{{}^3C_1 \cdot 3^2 + {}^3C_2 \cdot 3^1 + {}^3C_3}{6^3} = \frac{37}{216}$$

8. **Ans (B)**

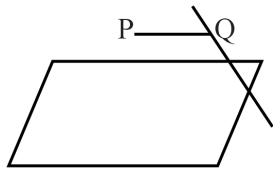
$$P(S) = \frac{1}{2} \times \frac{1}{6} + \frac{1}{2^2} \times \frac{2}{6^2} + \frac{1}{2^3} \times \frac{1}{6^3}$$

$$P(S) = \frac{144 + 24 + 1}{(12)^3} = \frac{169}{(12)^3}$$

$$P\left(\frac{B_2}{S}\right) = \frac{P(B_2 \cap S)}{P(S)} = \frac{2/(12)^2}{169/(12)^3} = \frac{24}{169}$$

9. **Ans (C)**

$$A(3, 2, 1), B(2, 3, 2); \vec{AB} = -\hat{i} + \hat{j} + \hat{k}$$



∴ lines are co-planar

$$\begin{vmatrix} -1 & 1 & 1 \\ 2 & 3 & \lambda \\ 3 & 2 & 3 \end{vmatrix} = 0 \Rightarrow \lambda = 4$$

Normal vector of plane containing the lines is

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 4 \\ 3 & 2 & 3 \end{vmatrix} = \hat{i} + 6\hat{j} - 5\hat{k}$$

$$\text{Equation of plane : } 1(x - 3) + 6(y - 2) - 5(z - 1) = 0$$

$$\Rightarrow x + 6y - 5z = 10$$

Now lines intersect at P(5, 5, 5)

$$\text{For plane } 3x + 2y - 2z + 15 = 0$$

$$\vec{n} = 3\hat{i} + 2\hat{j} - 2\hat{k}$$

$$\text{Let } Q(2t + 1, 4t + 3, 3t + 2)$$

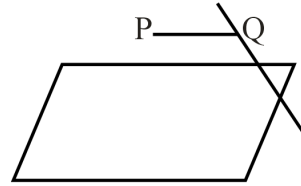
$$\therefore \vec{PQ} \cdot \vec{n} = 0 \Rightarrow t = \frac{5}{4}$$

$$\therefore \vec{PQ} = -\frac{3}{2}\hat{i} + 3\hat{j} + \frac{3}{4}\hat{k}$$

$$|\vec{PQ}| = \frac{3\sqrt{21}}{4}$$

10. **Ans (B)**

$$A(3, 2, 1), B(2, 3, 2); \vec{AB} = -\hat{i} + \hat{j} + \hat{k}$$



∴ lines are co-planar

$$\begin{vmatrix} -1 & 1 & 1 \\ 2 & 3 & \lambda \\ 3 & 2 & 3 \end{vmatrix} = 0 \Rightarrow \lambda = 4$$

Normal vector of plane containing the lines is

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 4 \\ 3 & 2 & 3 \end{vmatrix} = \hat{i} + 6\hat{j} - 5\hat{k}$$

$$\text{Equation of plane : } 1(x - 3) + 6(y - 2) - 5(z - 1) = 0$$

$$\Rightarrow x + 6y - 5z = 10$$

Now lines intersect at P(5, 5, 5)

$$\text{For plane } 3x + 2y - 2z + 15 = 0$$

$$\vec{n} = 3\hat{i} + 2\hat{j} - 2\hat{k}$$

$$\text{Let } Q(2t + 1, 4t + 3, 3t + 2)$$

$$\therefore \vec{PQ} \cdot \vec{n} = 0 \Rightarrow t = \frac{5}{4}$$

$$\therefore \vec{PQ} = -\frac{3}{2}\hat{i} + 3\hat{j} + \frac{3}{4}\hat{k}$$

$$|\vec{PQ}| = \frac{3\sqrt{21}}{4}$$

PART-3 : MATHEMATICS

SECTION-II (i)

1. **Ans (31.00)**

$$E_1 : \text{dot removed from odd face} \Rightarrow P(E_1) = \frac{9}{21}$$

$$E_2 : \text{dot removed from even face} \Rightarrow P(E_2) = \frac{12}{21}$$

E : die shows even number of dots.

$$P(E) = P(E \cap E_1) + P(E \cap E_2) = \frac{9}{21} \times \frac{2}{3} + \frac{12}{21} \times \frac{1}{3}$$

$$P(E) = \frac{10}{21} \quad \therefore P(\text{odd dots}) = \frac{11}{21}$$

2. Ans (320.00)

$$E_1 : \text{dot removed from odd face} \Rightarrow P(E_1) = \frac{9}{21}$$

$$E_2 : \text{dot removed from even face} \Rightarrow P(E_2) = \frac{12}{21}$$

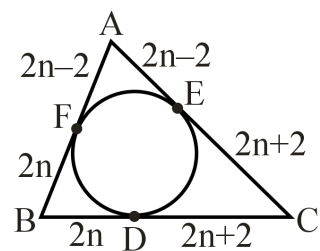
E : die shows even number of dots.

$$P(E) = P(E \cap E_1) + P(E \cap E_2) = \frac{9}{21} \times \frac{2}{3} + \frac{12}{21} \times \frac{1}{3}$$

$$P(E) = \frac{10}{21} \quad \therefore P(\text{odd dots}) = \frac{11}{21}$$

3. Ans (49.00)

$$a = 4n + 2, b = 4n, c = 4n - 2$$



$$S = 6n$$

$$\Delta = \sqrt{6n(2n+2)(2n-2)(2n)}$$

$$\Delta = 4n\sqrt{3(n^2-1)}$$

$$\text{Given } r = \frac{\Delta}{S} = \frac{2\sqrt{3(n^2-1)}}{3} = \frac{4\sqrt{6}}{3} \Rightarrow n = 3$$

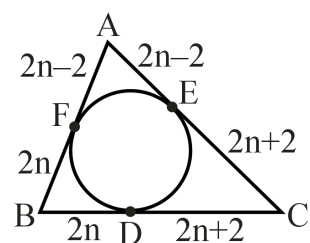
$$\therefore a = 14, b = 12, c = 10, \Delta = 24\sqrt{6}$$

$$R = \frac{abc}{4\Delta} = \frac{35}{2\sqrt{6}}$$

$$\ell_1^2 + \ell_2^2 + \ell_3^2 = \frac{3}{4}(a^2 + b^2 + c^2) = 330$$

4. Ans (330.00)

$$a = 4n + 2, b = 4n, c = 4n - 2$$



$$S = 6n$$

$$\Delta = \sqrt{6n(2n+2)(2n-2)(2n)}$$

$$\Delta = 4n\sqrt{3(n^2-1)}$$

$$\text{Given } r = \frac{\Delta}{S} = \frac{2\sqrt{3(n^2-1)}}{3} = \frac{4\sqrt{6}}{3} \Rightarrow n = 3$$

$$\therefore a = 14, b = 12, c = 10, \Delta = 24\sqrt{6}$$

$$R = \frac{abc}{4\Delta} = \frac{35}{2\sqrt{6}}$$

$$\ell_1^2 + \ell_2^2 + \ell_3^2 = \frac{3}{4}(a^2 + b^2 + c^2) = 330$$

5. Ans (0.25)

$$\vec{a} \times \vec{b} + \vec{b} \times \vec{c} = p\vec{a} + q\vec{b} + r\vec{c} \text{ taking dot product}$$

with $\vec{a}$

$$0 + [abc] = p\vec{a} \cdot \vec{a} + q\vec{a} \cdot \vec{b} + r\vec{a} \cdot \vec{c}$$

$$[abc] = p + \frac{q}{2} + \frac{r}{2} \quad \dots(1)$$

Similarly taking dot product with $\vec{b}$

$$0 = \frac{p}{2} + q + \frac{r}{2} \quad \dots(2)$$

and taking dot product with $\vec{c}$

$$[\vec{a} \quad \vec{b} \quad \vec{c}] = \frac{p}{2} + \frac{q}{2} + r \quad \dots(3)$$

From equation (1) and (3), we get $p = r$

From equation (2) $p = -q$

$$\Rightarrow [abc] = p - \frac{p}{2} + \frac{p}{2} = p$$

But we know that

$$[\vec{a} \quad \vec{b} \quad \vec{c}]^2 = \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix} = \begin{vmatrix} 1 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 1 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 1 \end{vmatrix}$$

$$\Rightarrow [\vec{a} \quad \vec{b} \quad \vec{c}]^2 = \frac{1}{2}$$

$$\Rightarrow [\vec{a} \quad \vec{b} \quad \vec{c}]^4 = \frac{1}{4} = 0.25$$

$$p = \frac{1}{\sqrt{2}}, q = -\frac{1}{\sqrt{2}}, r = \frac{1}{\sqrt{2}}$$

$$\text{Now, } 20p^2 + 21q^2 - 22r^2 = \frac{20}{2} + \frac{21}{2} - \frac{22}{2} = 9.50$$

6. Ans (9.50)

$\vec{a} \times \vec{b} + \vec{b} \times \vec{c} = p\vec{a} + q\vec{b} + r\vec{c}$ taking dot product

with $\vec{a}$

$$0 + [abc] = p\vec{a} \cdot \vec{a} + q\vec{a} \cdot \vec{b} + r\vec{a} \cdot \vec{c}$$

$$[abc] = p + \frac{q}{2} + \frac{r}{2} \quad \dots(1)$$

Similarly taking dot product with $\vec{b}$

$$0 = \frac{p}{2} + q + \frac{r}{2} \quad \dots(2)$$

and taking dot product with $\vec{c}$

$$[\vec{a} \quad \vec{b} \quad \vec{c}] = \frac{p}{2} + \frac{q}{2} + r \quad \dots(3)$$

From equation (1) and (3), we get $p = r$

From equation (2) $p = -q$

$$\Rightarrow [abc] = p - \frac{p}{2} + \frac{p}{2} = p$$

But we know that

$$[\vec{a} \quad \vec{b} \quad \vec{c}]^2 = \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix} = \begin{vmatrix} 1 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 1 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 1 \end{vmatrix}$$

$$\Rightarrow [\vec{a} \quad \vec{b} \quad \vec{c}]^2 = \frac{1}{2}$$

$$\Rightarrow [\vec{a} \quad \vec{b} \quad \vec{c}]^4 = \frac{1}{4} = 0.25$$

$$p = \frac{1}{\sqrt{2}}, q = -\frac{1}{\sqrt{2}}, r = \frac{1}{\sqrt{2}}$$

$$\text{Now, } 20p^2 + 21q^2 - 22r^2 = \frac{20}{2} + \frac{21}{2} - \frac{22}{2} = 9.50$$

PART-3 : MATHEMATICS

SECTION-II (ii)

7. Ans (7)

Circumcentre of ΔABC is at origin.

$\therefore$ Orthocentre of ΔDEF is at origin.

$$|\overrightarrow{PQ}| = 7$$

8. Ans (135)

Let $S = \{a_1, a_2, a_3, a_4, a_5\}$

For each a_i , four cases are possible.

(i) $a_i \in A$ and $a_i \in B$

(ii) $a_i \in A$ but $a_i \notin B$

(iii) $a_i \notin A$ and $a_i \in B$

(iv) $a_i \notin A$ and $a_i \notin B$

Thus total number of ways of choosing A and B is 4^5

for $P \cap Q$ to contain exactly 2 elements, no. of

$$\text{ways} = {}^5C_2(3)^3$$

$$P = \frac{{}^5C_2(3)^3}{4^5} = \frac{135}{512}$$

9. Ans (2)

$$\vec{r}_1 + 2\vec{r}_2 = \lambda(2\vec{r}_1 + \vec{r}_2)$$

$$(p+2)\vec{a} + (q+2p)\vec{b} + (1+2q)\vec{c}$$

$$= \lambda[(2p+1)\vec{a} + (2q+p)\vec{b} + (2+q)\vec{c}]$$

$$\frac{p+2}{2p+1} = \frac{q+2p}{p+2q} = \frac{2q+1}{q+2} = \lambda$$

$$\Rightarrow p = q = 1$$